

In [4]: `import pandas as pd`

```
# Loading the dataset
df = pd.read_csv(r"Dataset/Heart Disease UCI.csv")
df.head(10)
```

Out[4]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	69	1	0	160	234	1	2	131	0	0.1	1	1	0
1	69	0	0	140	239	0	0	151	0	1.8	0	2	0
2	66	0	0	150	226	0	0	114	0	2.6	2	0	0
3	65	1	0	138	282	1	2	174	0	1.4	1	1	0
4	64	1	0	110	211	0	2	144	1	1.8	1	0	0
5	64	1	0	170	227	0	2	155	0	0.6	1	0	2
6	63	1	0	145	233	1	2	150	0	2.3	2	0	1
7	61	1	0	134	234	0	0	145	0	2.6	1	2	0
8	60	0	0	150	240	0	0	171	0	0.9	0	0	0
9	59	1	0	178	270	0	2	145	0	4.2	2	0	2



In [14]: `# Checks null values in the dataset`

```
df.isna().value_counts()
# df.isna().sum()
```

Out[14]:

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	condition
False	False	False	False	False	False	False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False	False	False	False	False	False	False
Name: count, dtype: int64													

In [15]: `# Validating datatypes`

```
df.dtypes
```

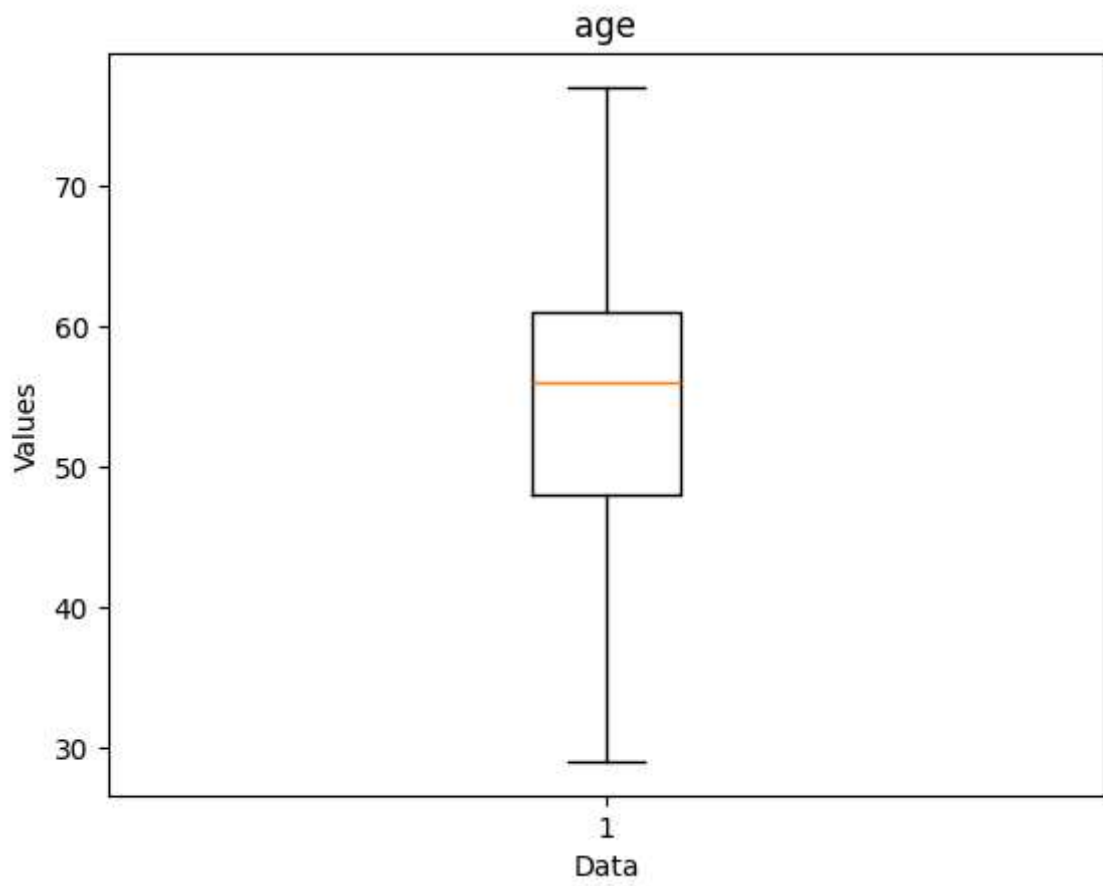
Out[15]:

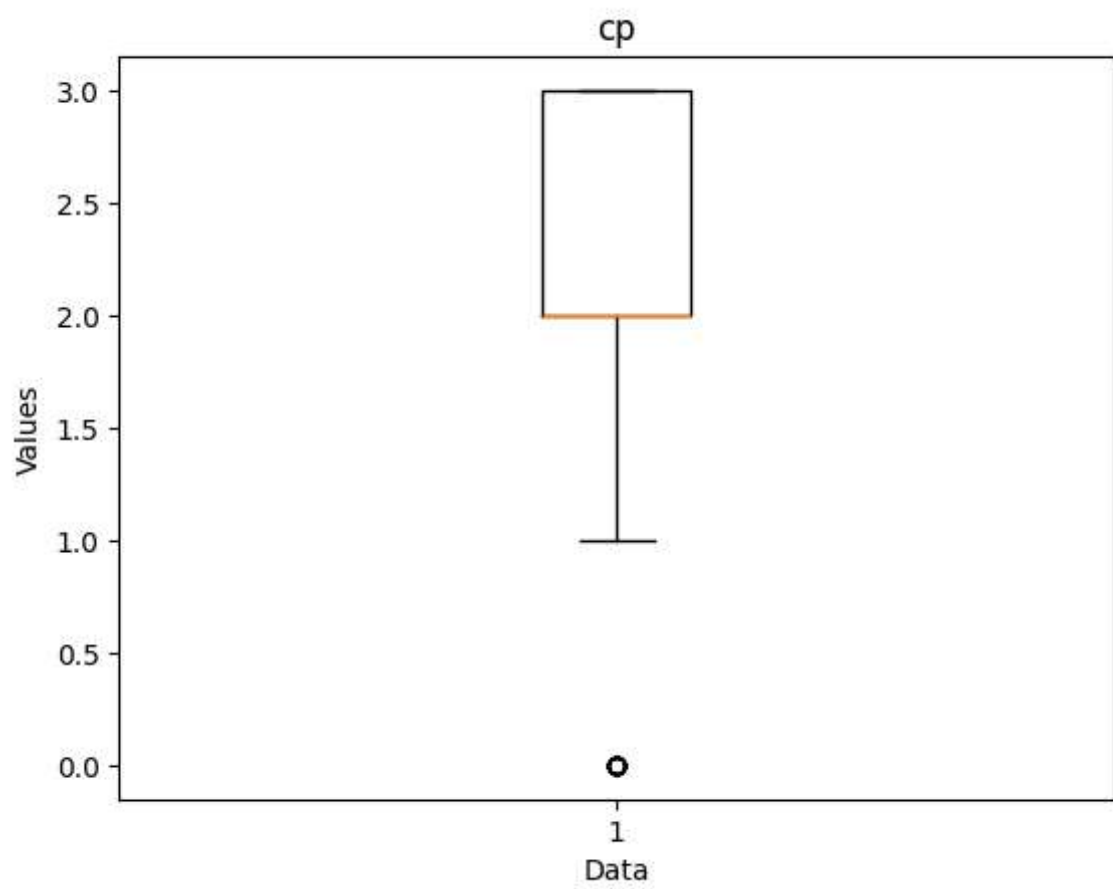
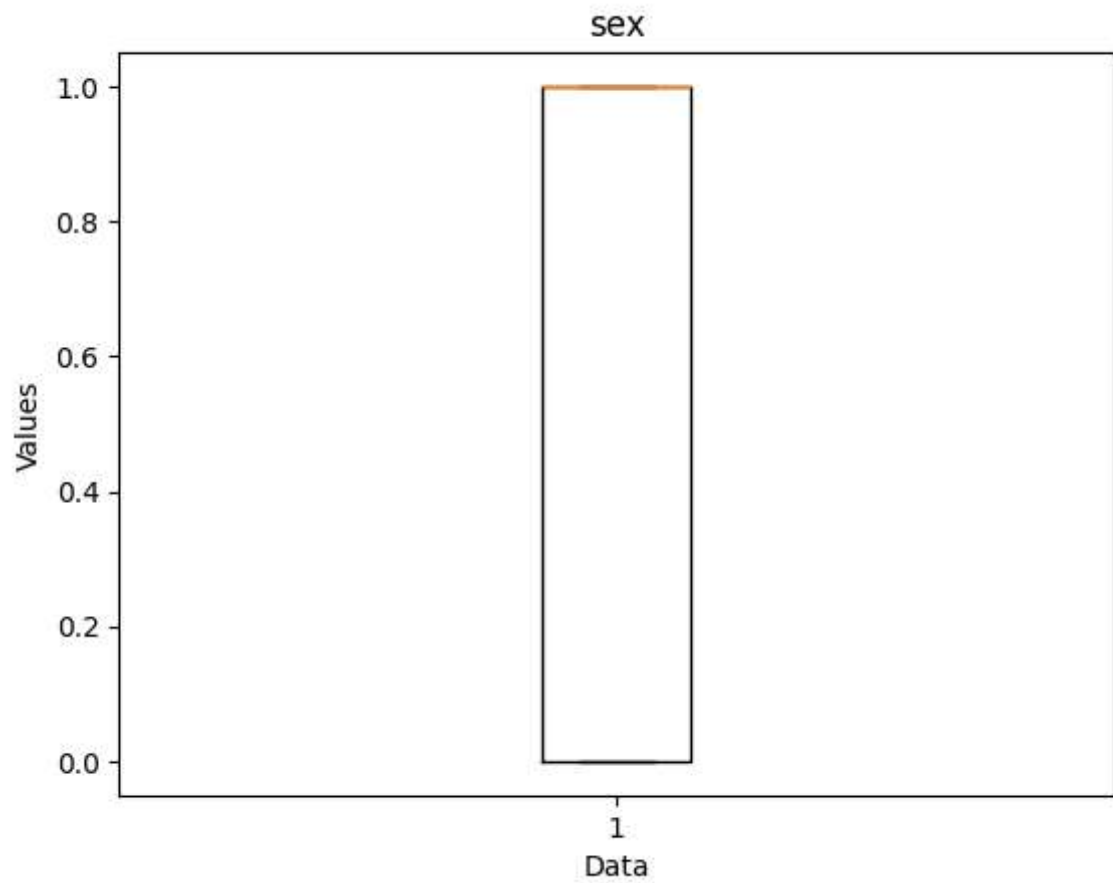
```
age      int64
sex      int64
cp       int64
trestbps int64
chol     int64
fbs      int64
restecg  int64
thalach  int64
exang    int64
oldpeak  float64
slope    int64
ca       int64
thal     int64
condition int64
dtype: object
```

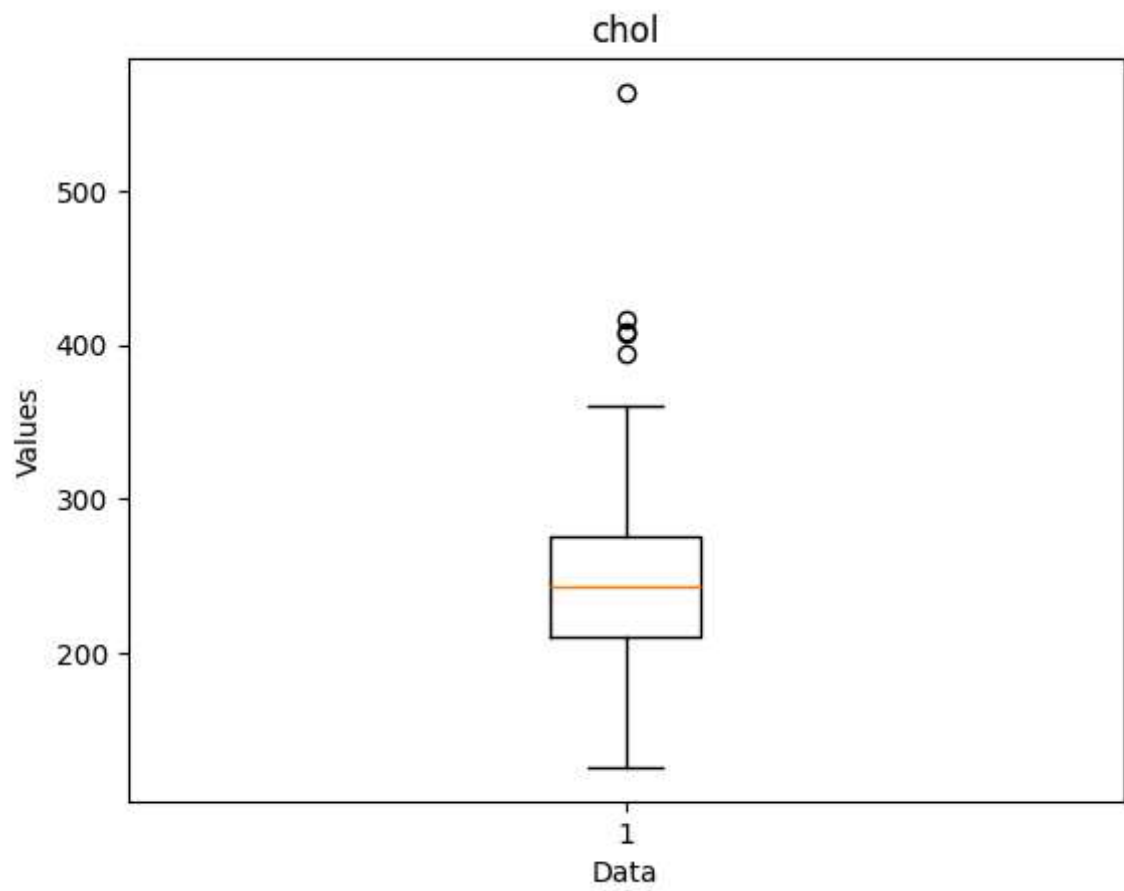
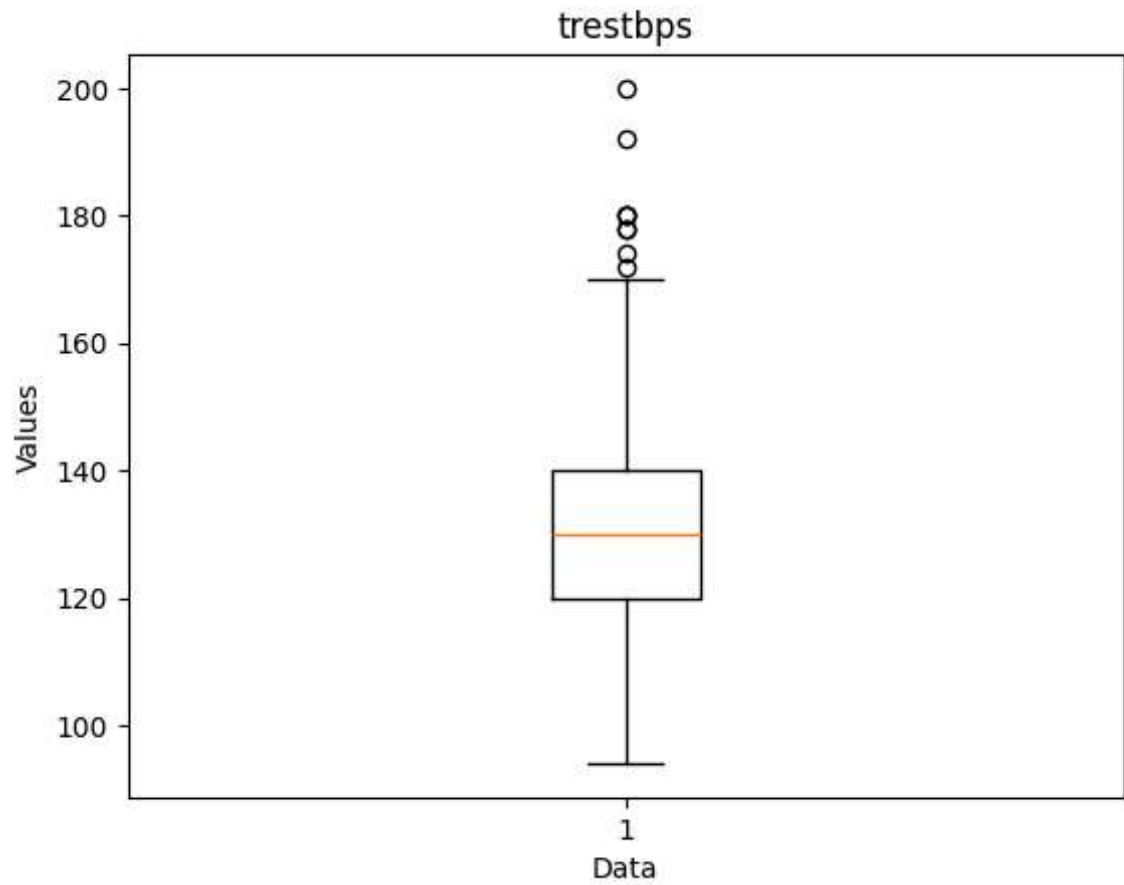
In [9]: `# Detect outliers`

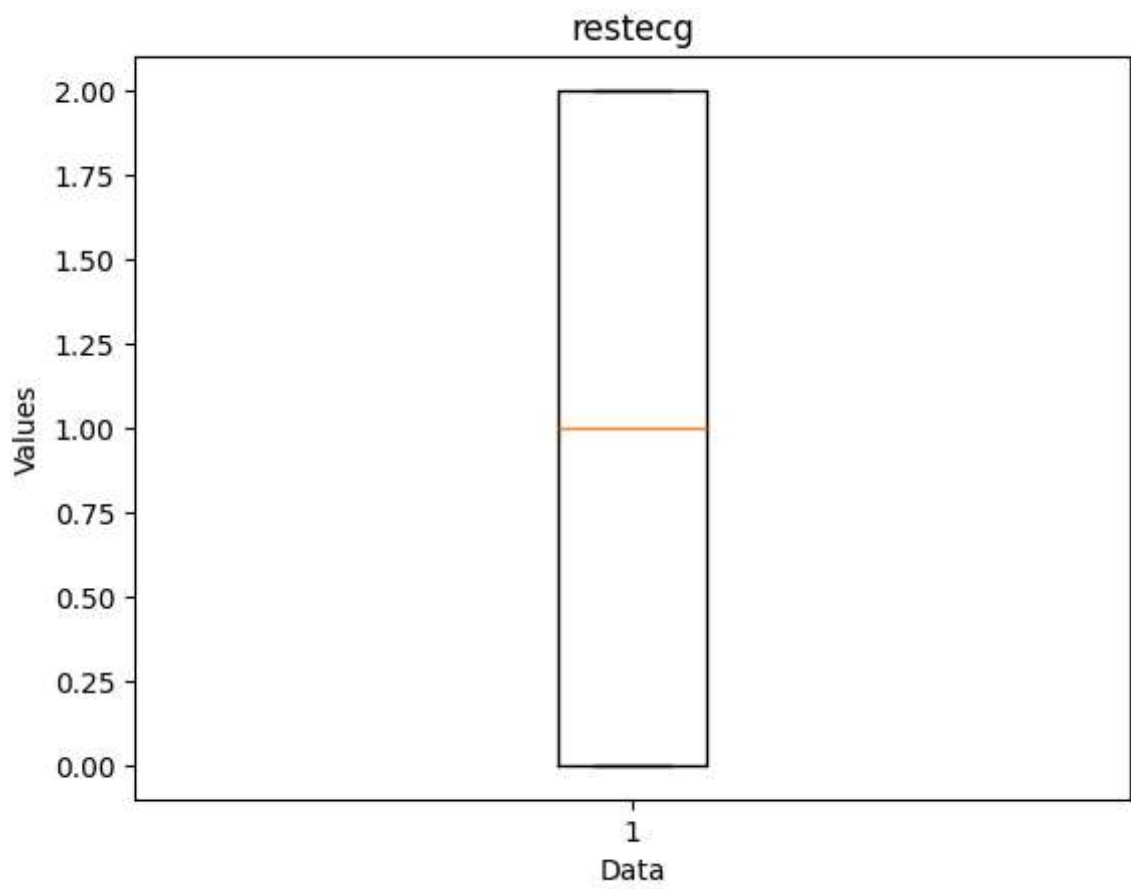
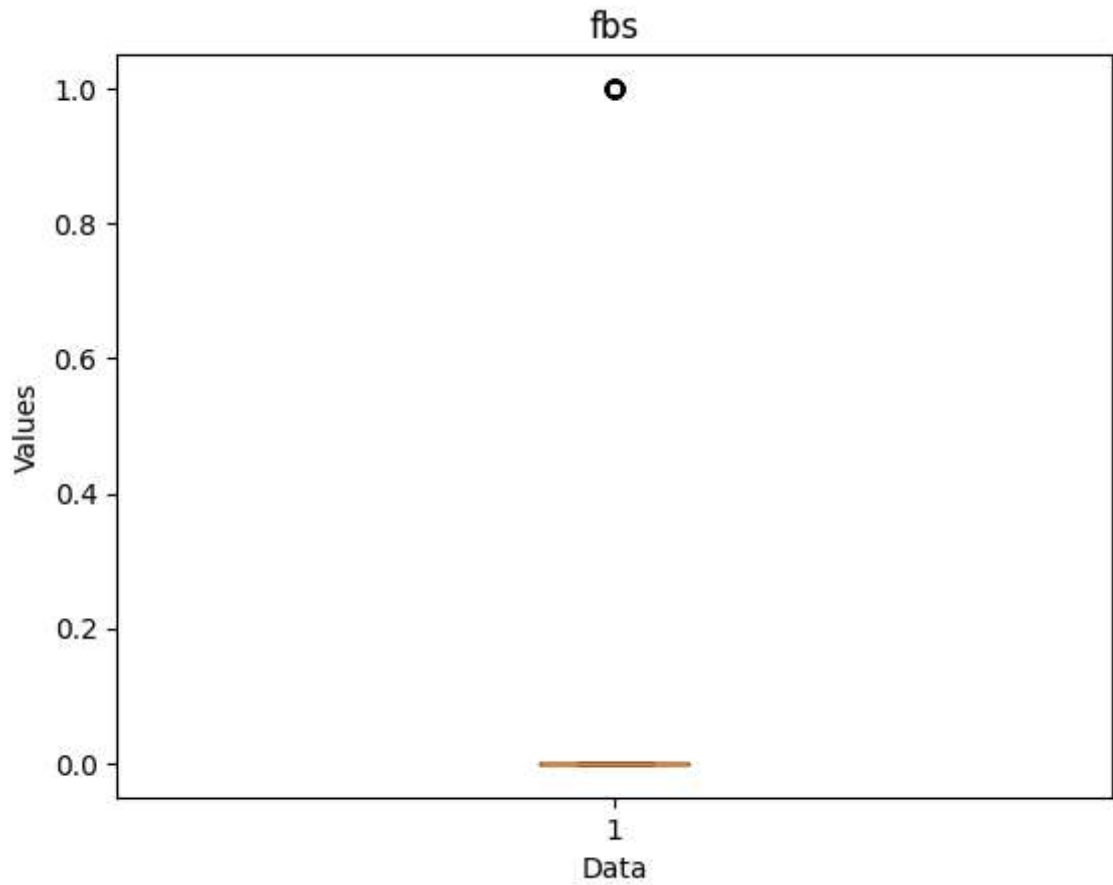
```
import matplotlib.pyplot as plt
```

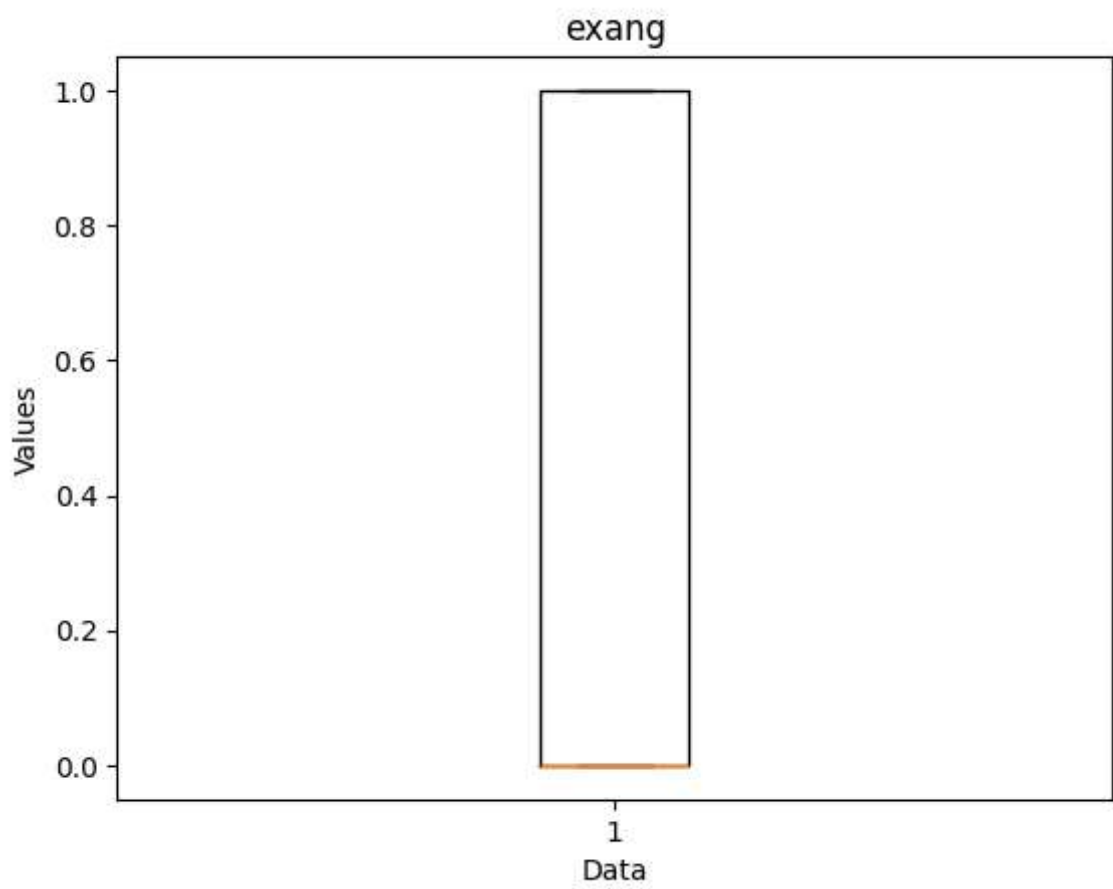
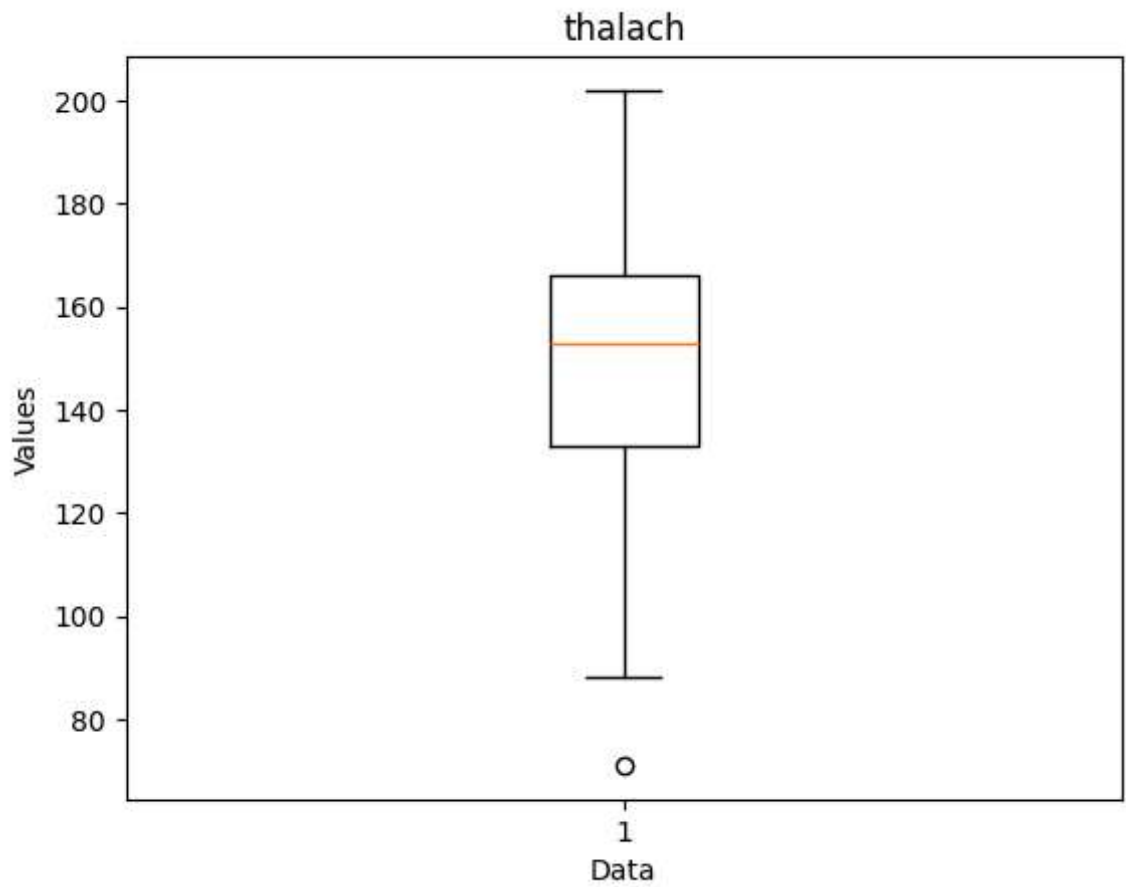
```
for column in df.columns:  
    data = list(df[column])  
    plt.boxplot(data)  
    plt.title(column)  
    plt.xlabel('Data')  
    plt.ylabel('Values')  
  
plt.show()
```

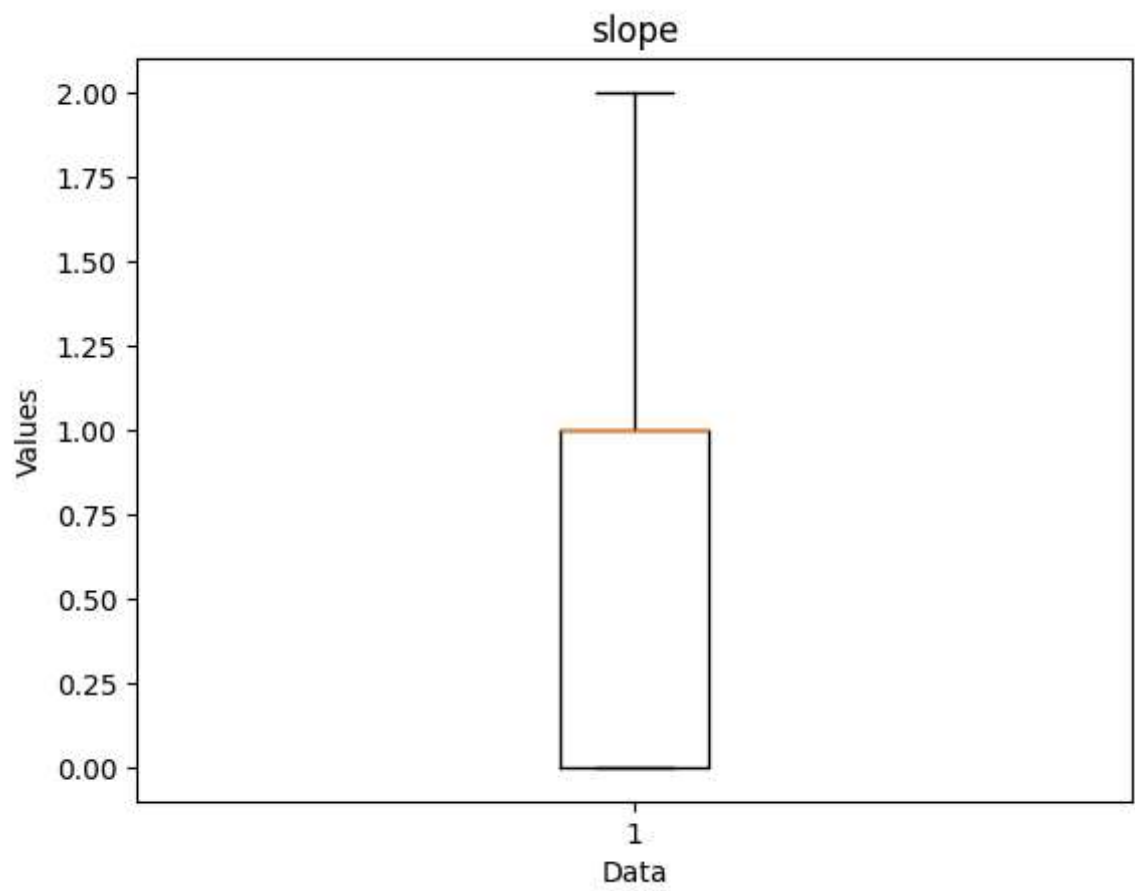
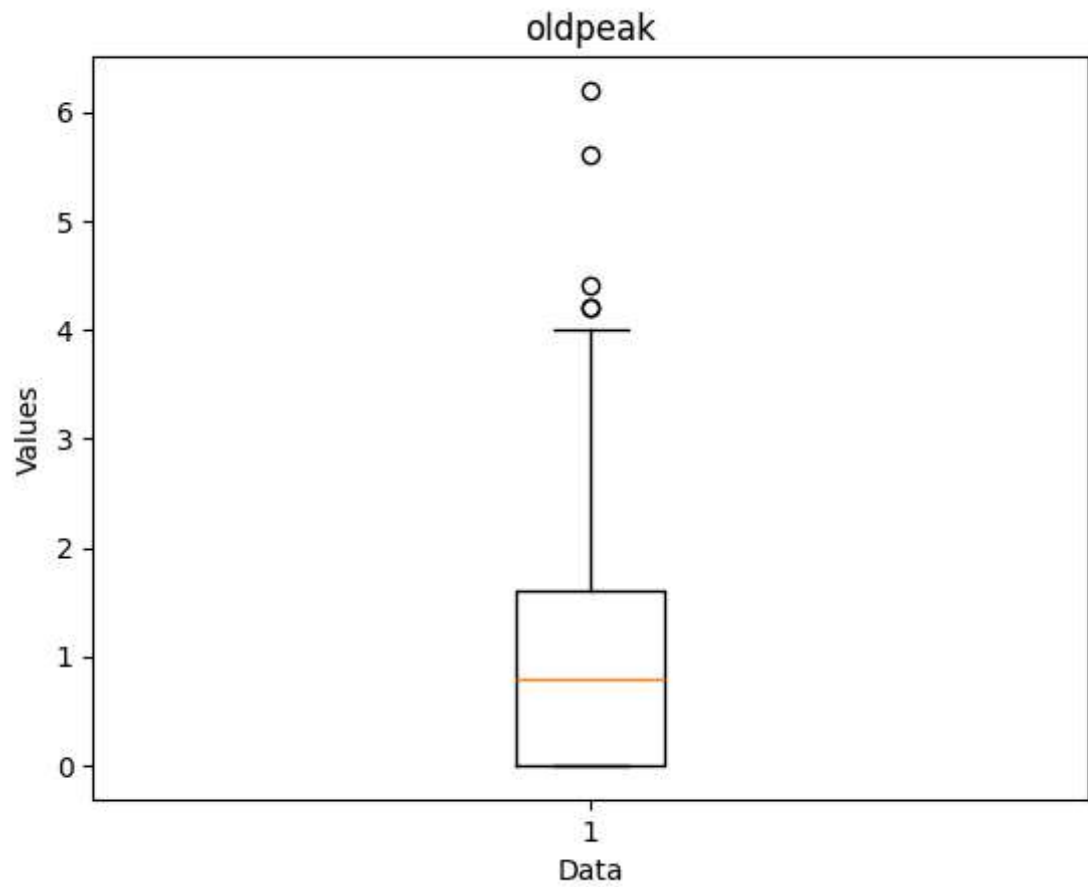


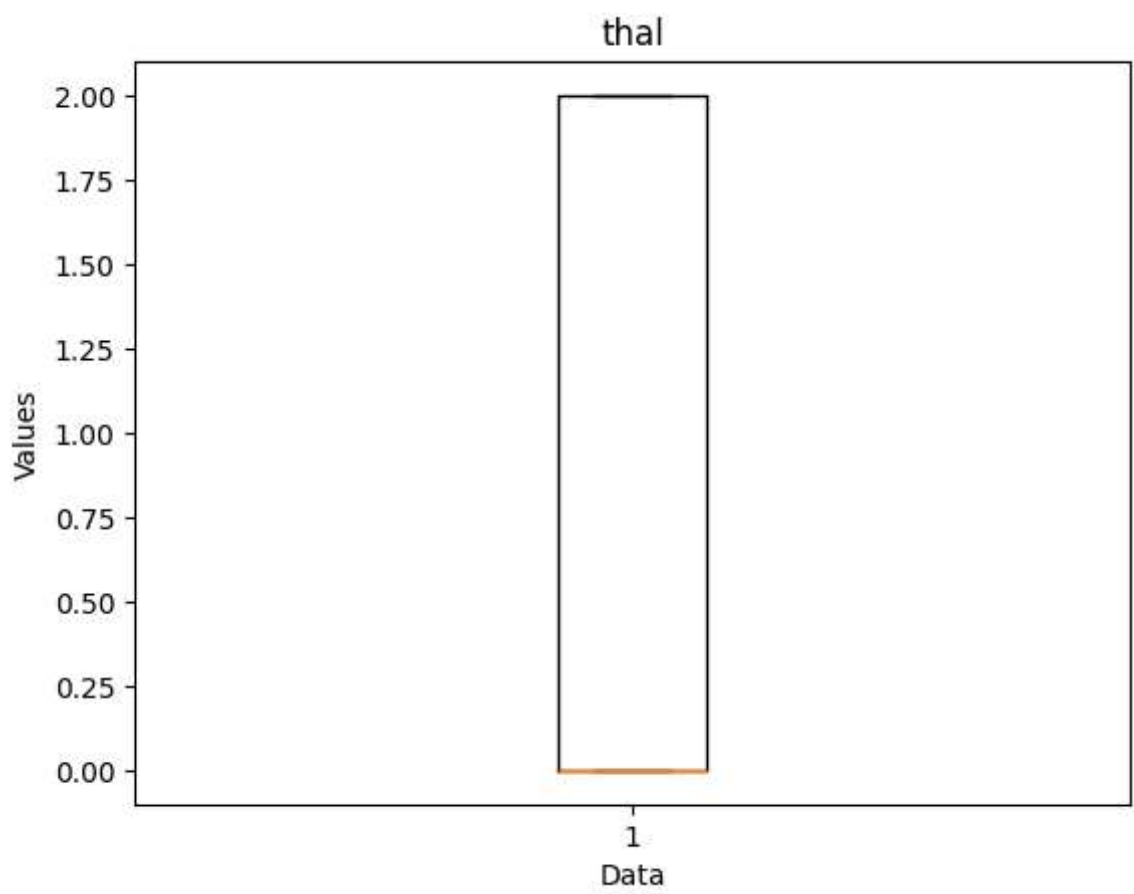
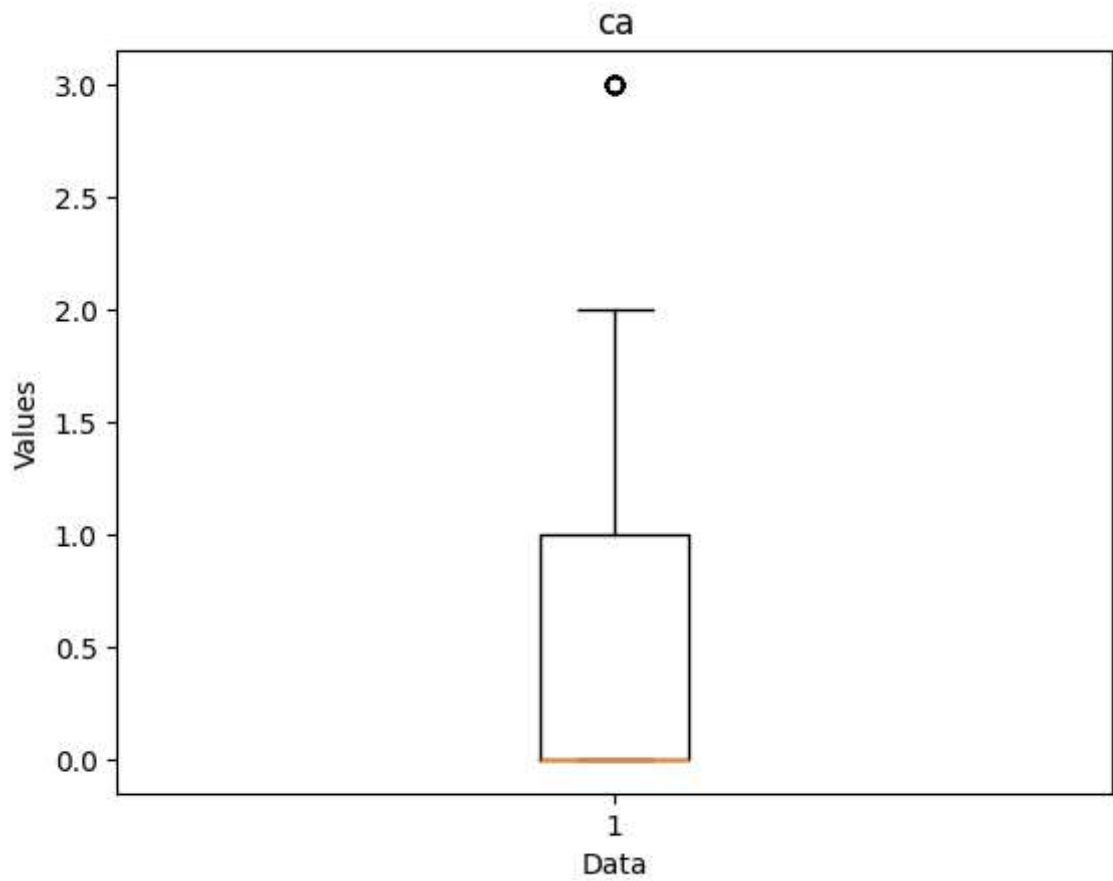


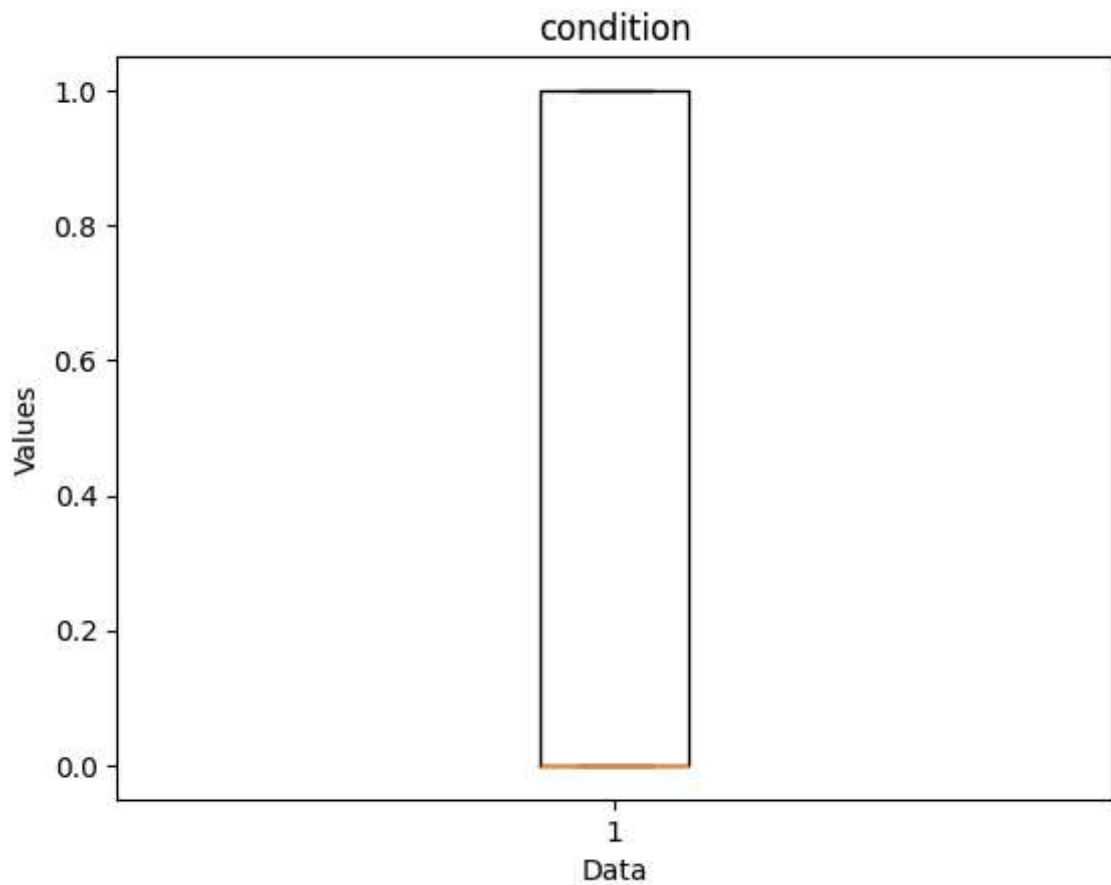












```
In [25]: # Summary statistics
df.describe()
```

```
Out[25]:
```

	age	sex	cp	trestbps	chol	fbs	restecg
count	297.000000	297.000000	297.000000	297.000000	297.000000	297.000000	297.000000
mean	54.542088	0.676768	2.158249	131.693603	247.350168	0.144781	0.996633
std	9.049736	0.468500	0.964859	17.762806	51.997583	0.352474	0.994915
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	48.000000	0.000000	2.000000	120.000000	211.000000	0.000000	0.000000
50%	56.000000	1.000000	2.000000	130.000000	243.000000	0.000000	1.000000
75%	61.000000	1.000000	3.000000	140.000000	276.000000	0.000000	2.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

```
In [ ]:
```

```
In [15]: maleCount = df[df['sex'] == 1].sex.count()
femaleCount = df[df['sex'] == 0].sex.count()
print(maleCount, femaleCount)
# print(df['sex'].value_counts())
```

```

201 96
sex
1    201
0     96
Name: count, dtype: int64

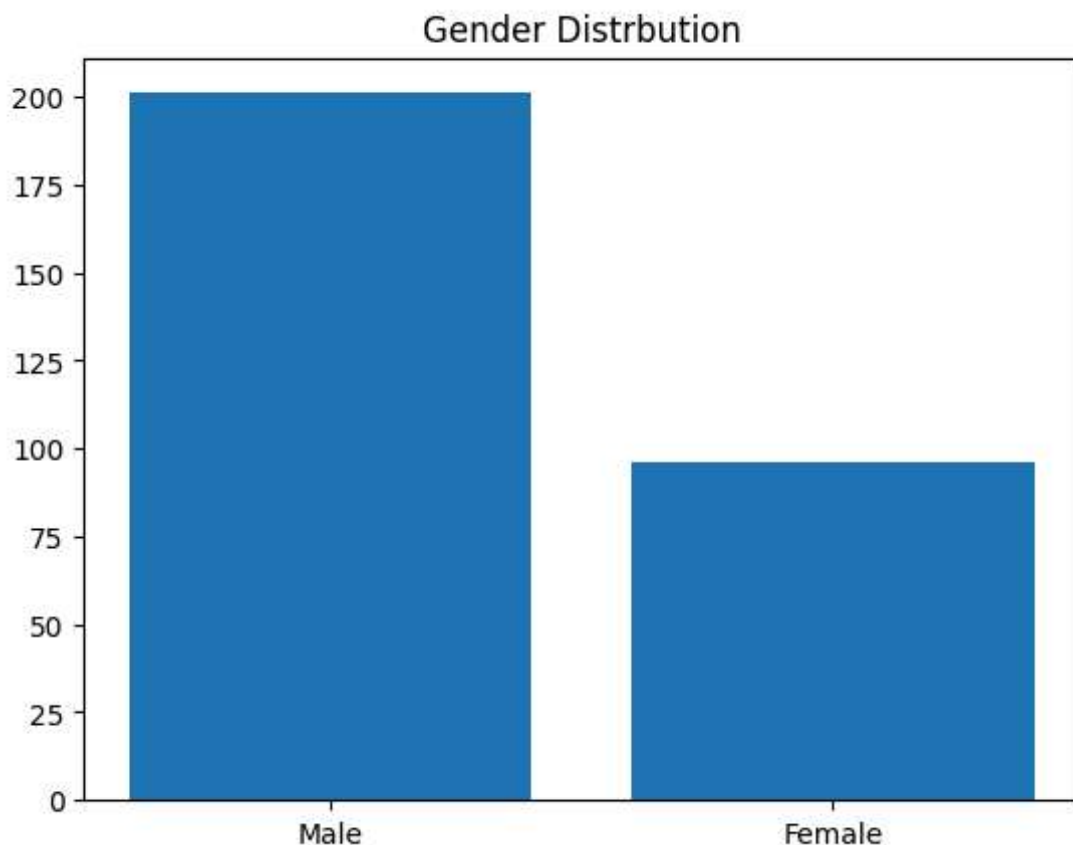
```

```
In [34]: import numpy as np
```

```
In [43]: # print(np.percentile(df['trestbps'],0))
```

```
94.0
```

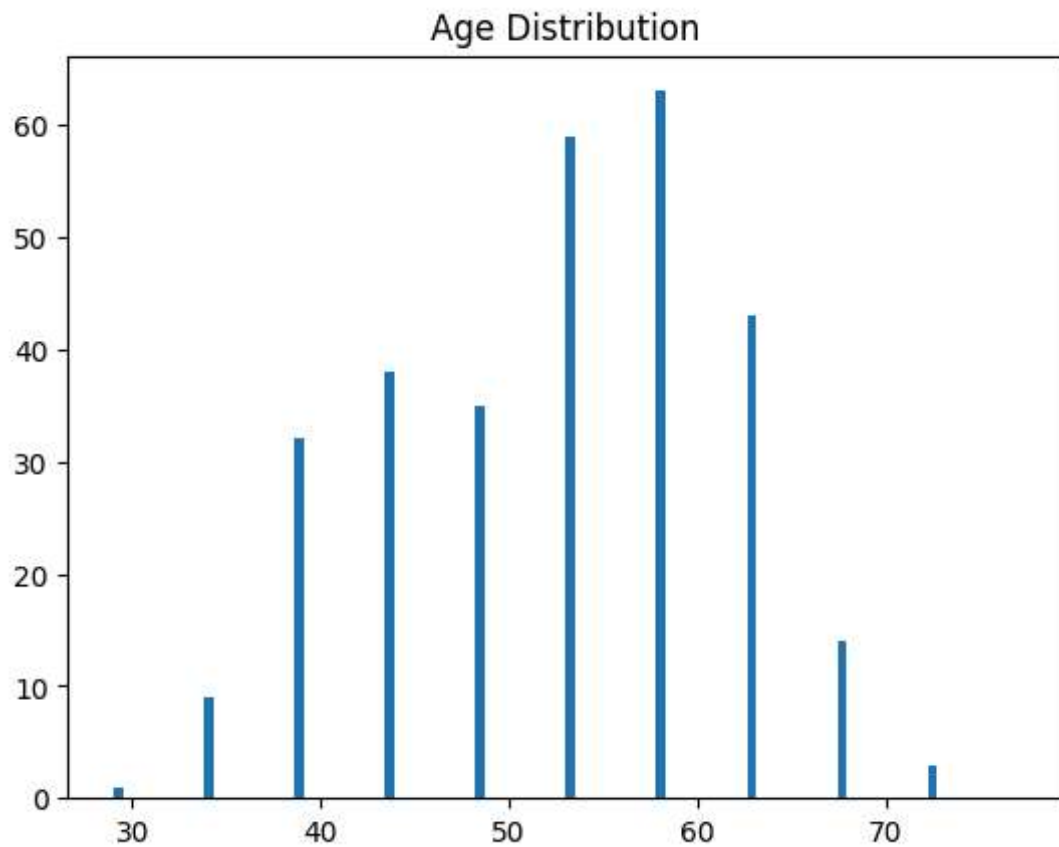
```
In [106]: plt.bar(['Male', 'Female'], [maleCount, femaleCount])
plt.title('Gender Distribution')
plt.show()
```



```
In [114]: minAge = df['age'].min()
maxAge = df['age'].max()
meanAge = df['age'].mean()
medianAge = df['age'].median()
print(maxAge, minAge, meanAge, medianAge)
```

```
77 29 54.54208754208754 56.0
```

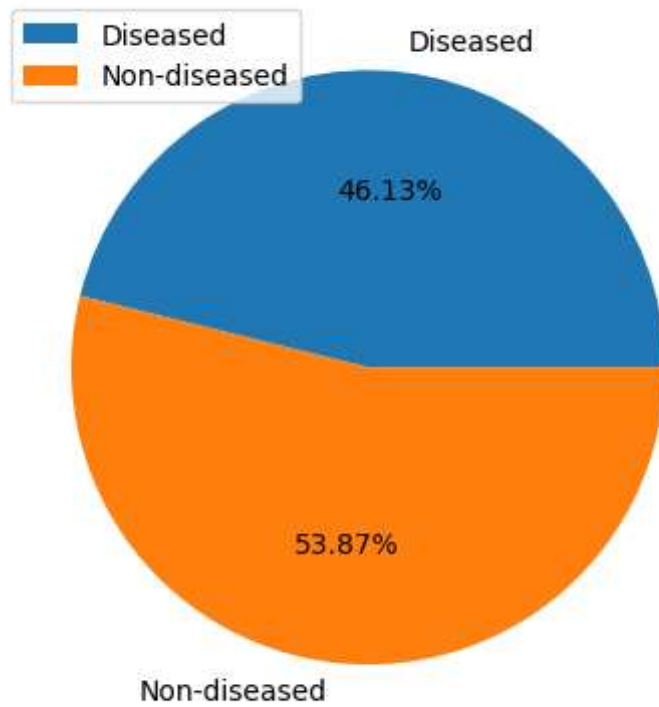
```
In [119]: plt.hist(df['age'], width = 0.5)
plt.title('Age Distribution')
plt.show()
```



```
In [125]: ''' Target Variable Analysis '''  
  
diseased = df[df['condition'] == 1].condition.count()  
nondiseased = df[df['condition'] == 0].condition.count()  
  
print(diseased, nondiseased)
```

137 160

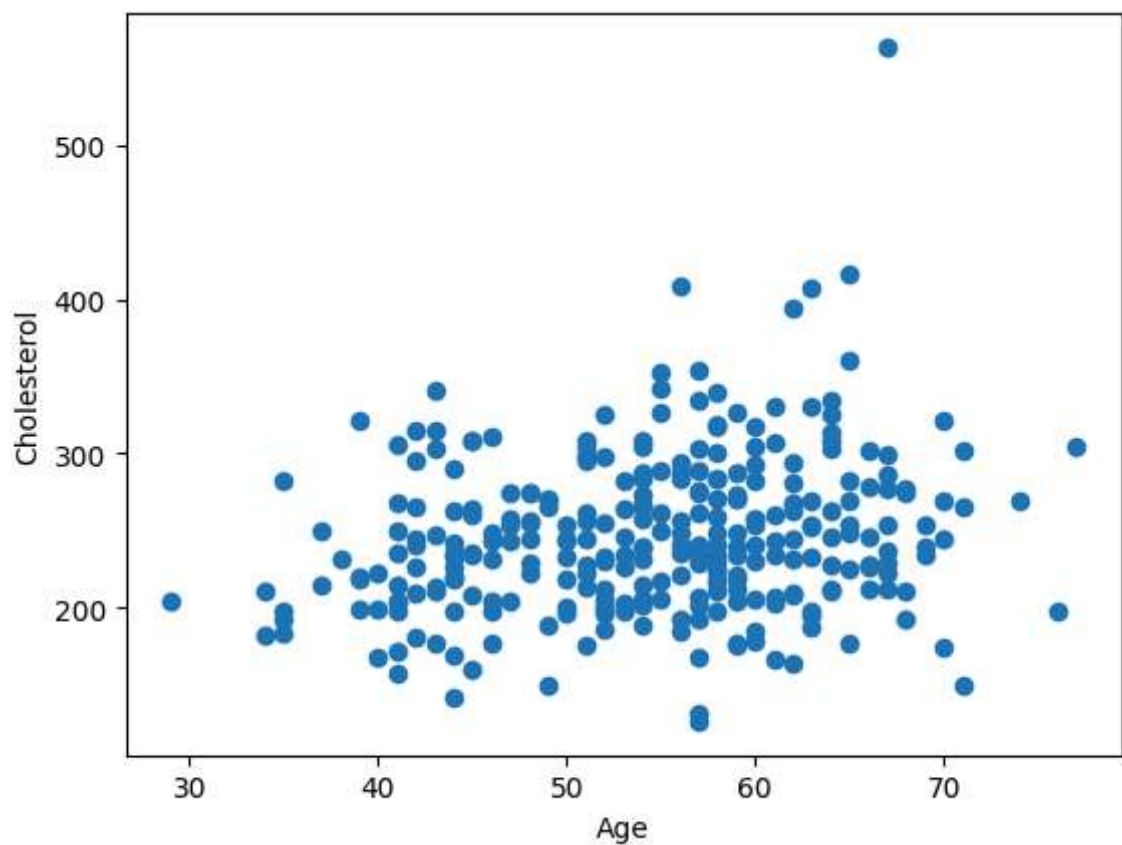
```
In [131]: plt.pie([diseased, nondiseased] , labels = ['Diseased', 'Non-diseased'], autopct =  
plt.legend()  
plt.show()
```



```
In [28]: df.corr()  
df['age'].corr(df['chol'])
```

```
Out[28]: 0.20264354584662683
```

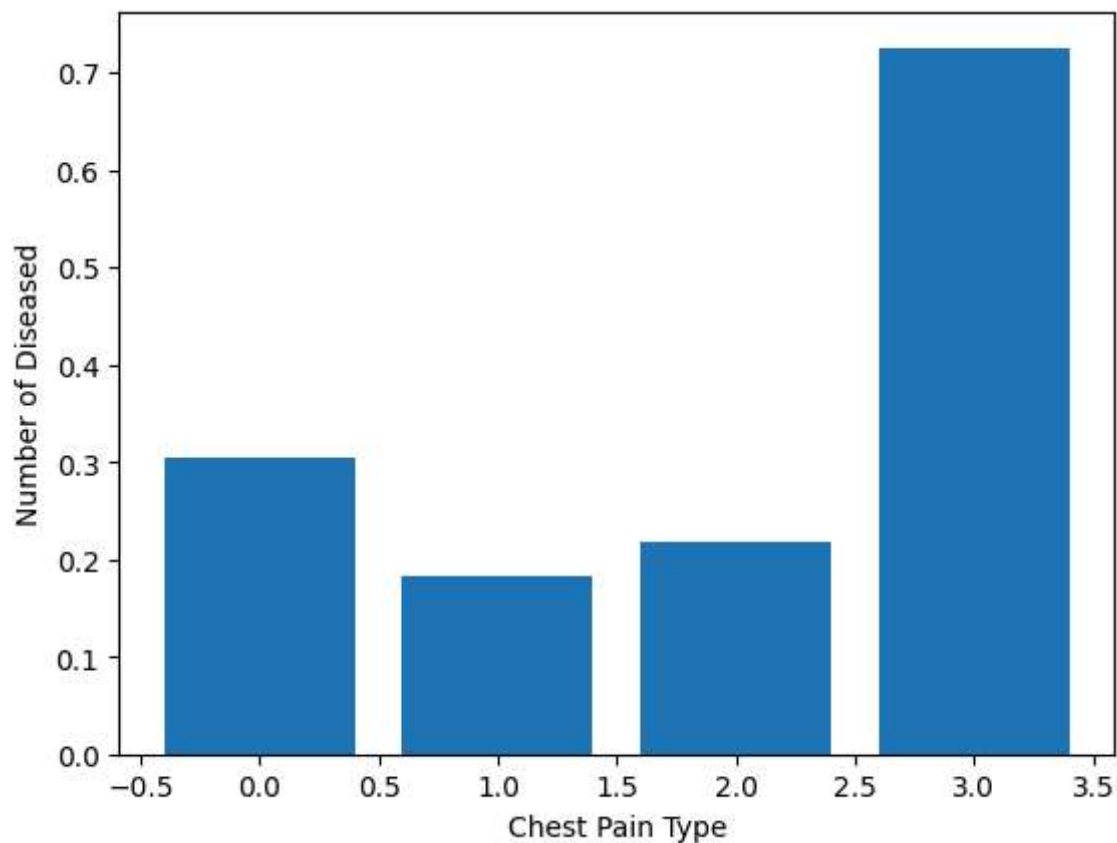
```
In [139]: plt.scatter(df['age'], df['chol'])  
plt.xlabel('Age')  
plt.ylabel('Cholesterol')  
plt.show()
```



```
In [23]: cpGroup = df.groupby('cp')
cpGroupCounts = cpGroup.condition.sum()
```

```
cp
0 ..... 7
1 ..... 9
2 ..... 18
3 ..... 103
Name: condition, dtype: int64
cp
0 ..... 23
1 ..... 49
2 ..... 83
3 ..... 142
Name: condition, dtype: int64
```

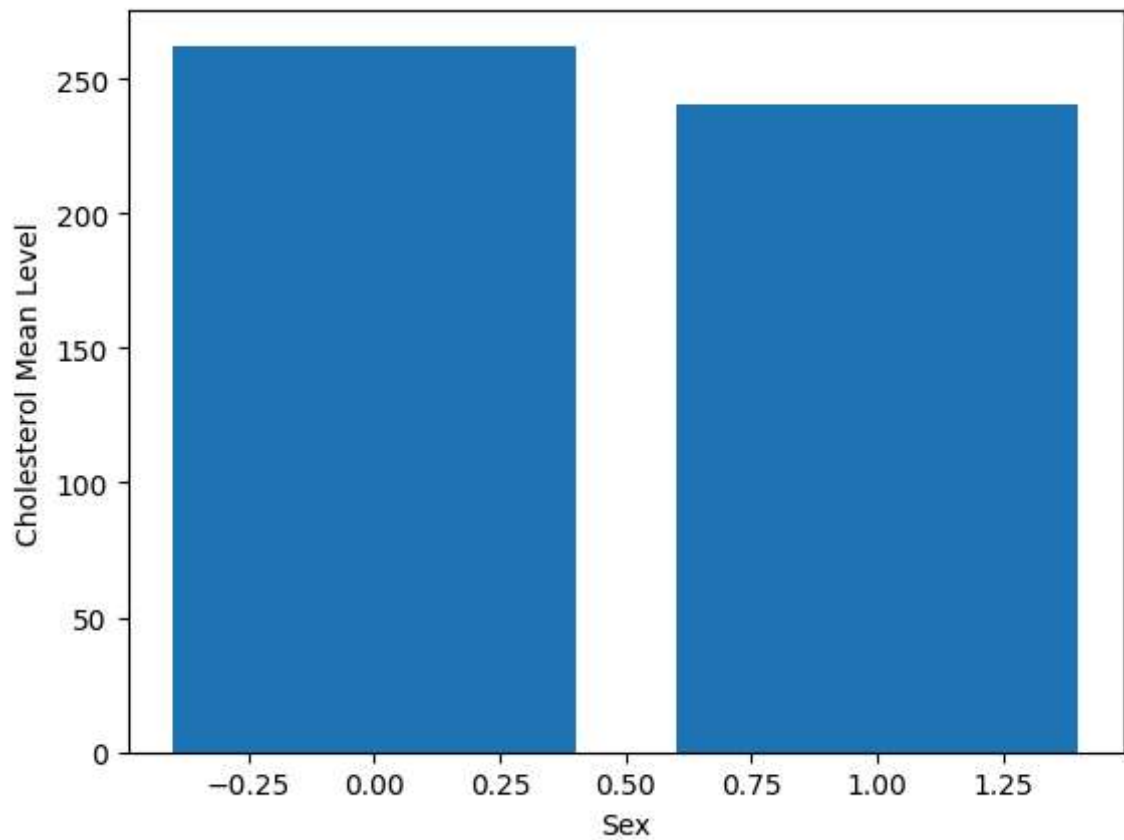
```
In [24]: plt.bar([ i for i in cpGroupCounts.index], cpGroupCounts/cpGroup.condition.count())
plt.xlabel('Chest Pain Type')
plt.ylabel('Number of Diseased')
plt.show()
```



```
In [ ]:
```

```
In [20]: sexGroup = df.groupby('sex')
sexGroupMeanChol = sexGroup.mean()['chol']
```

```
In [165]: plt.bar(sexGroupMeanChol.index, sexGroupMeanChol)
plt.xlabel("Sex")
plt.ylabel("Cholesterol Mean Level")
plt.show()
```



```
In [170]: avgBP = df['trestbps'].mean()
avgBP
```

```
Out[170]: 131.69360269360268
```

```
In [176]: highBP = df[df['trestbps'] > 140]
highBP
```

```
Out[176]:
```

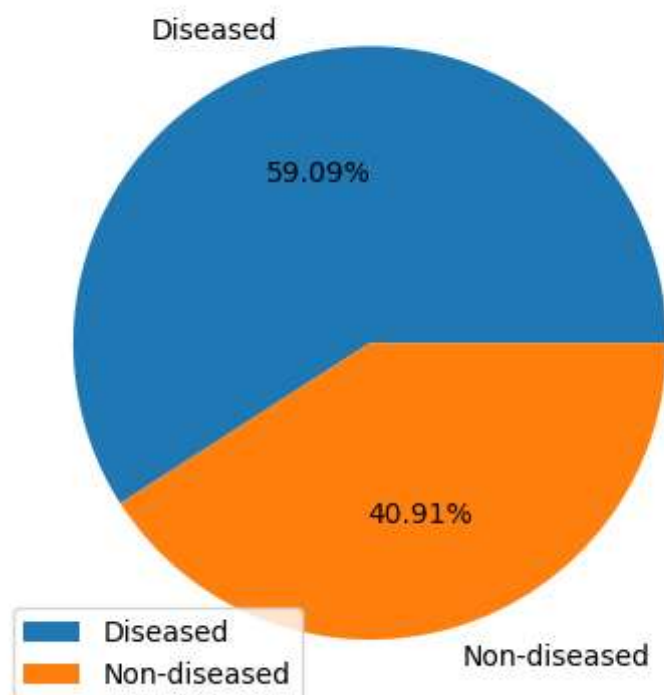
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	69	1	0	160	234	1	2	131	0	0.1	1	1	
2	66	0	0	150	226	0	0	114	0	2.6	2	0	
5	64	1	0	170	227	0	2	155	0	0.6	1	0	
6	63	1	0	145	233	1	2	150	0	2.3	2	0	
8	60	0	0	150	240	0	0	171	0	0.9	0	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
263	50	1	3	150	243	0	2	128	0	2.6	1	0	
264	50	1	3	144	200	0	2	126	1	0.9	1	0	
277	45	1	3	142	309	0	2	147	1	0.0	1	3	
285	43	1	3	150	247	0	0	171	0	1.5	0	0	
292	40	1	3	152	223	0	0	181	0	0.0	0	0	

66 rows × 14 columns



```
In [194]: diseased = highBP.condition.sum()
nondiseased = highBP.condition.count() - diseased
plt.pie([diseased, nondiseased], labels = ['Diseased', 'Non-diseased'], autopct = '%.09f%%')
plt.legend()
plt.title("Disease presence for patients with BP > 140")
plt.show()
```

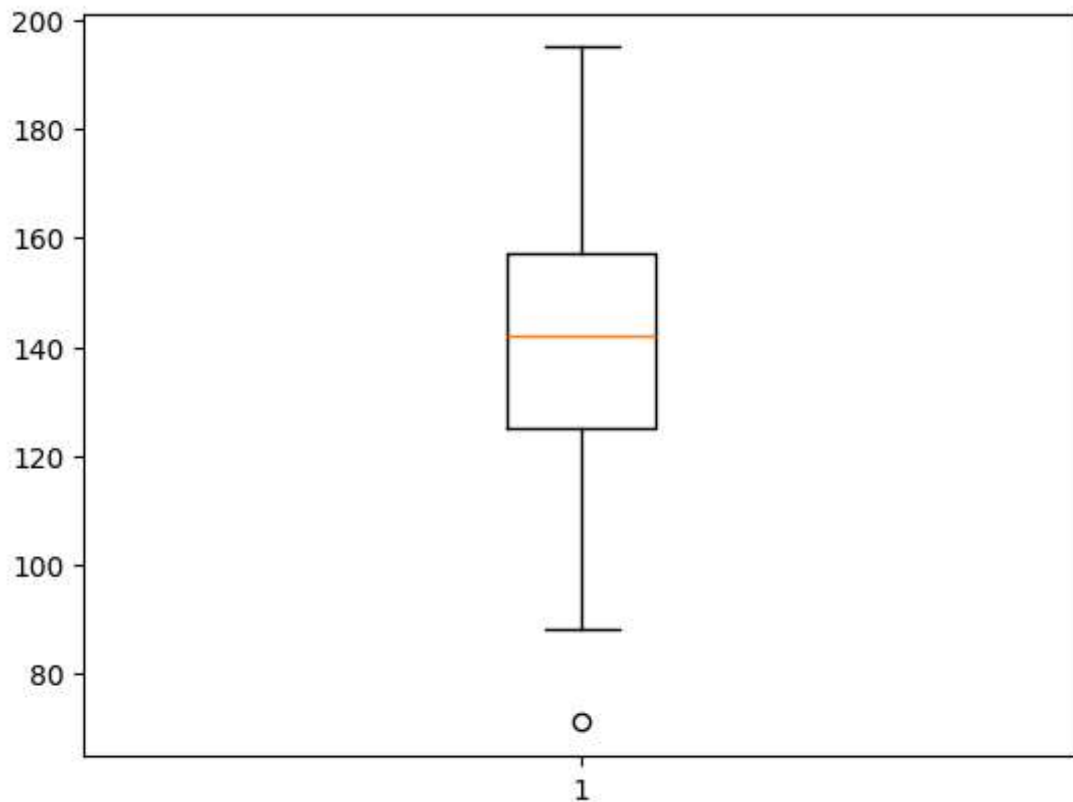
Disease presence for patients with BP > 140



```
In [216]: illThala = df[df['condition'] == 1].thalach
nonIllThala = df[df['condition'] == 0].thalach
print(illThala.mean(), nonIllThala.mean())
```

139.1094890510949 158.58125

```
In [223]: plt.boxplot(list(illThala))
plt.show()
```

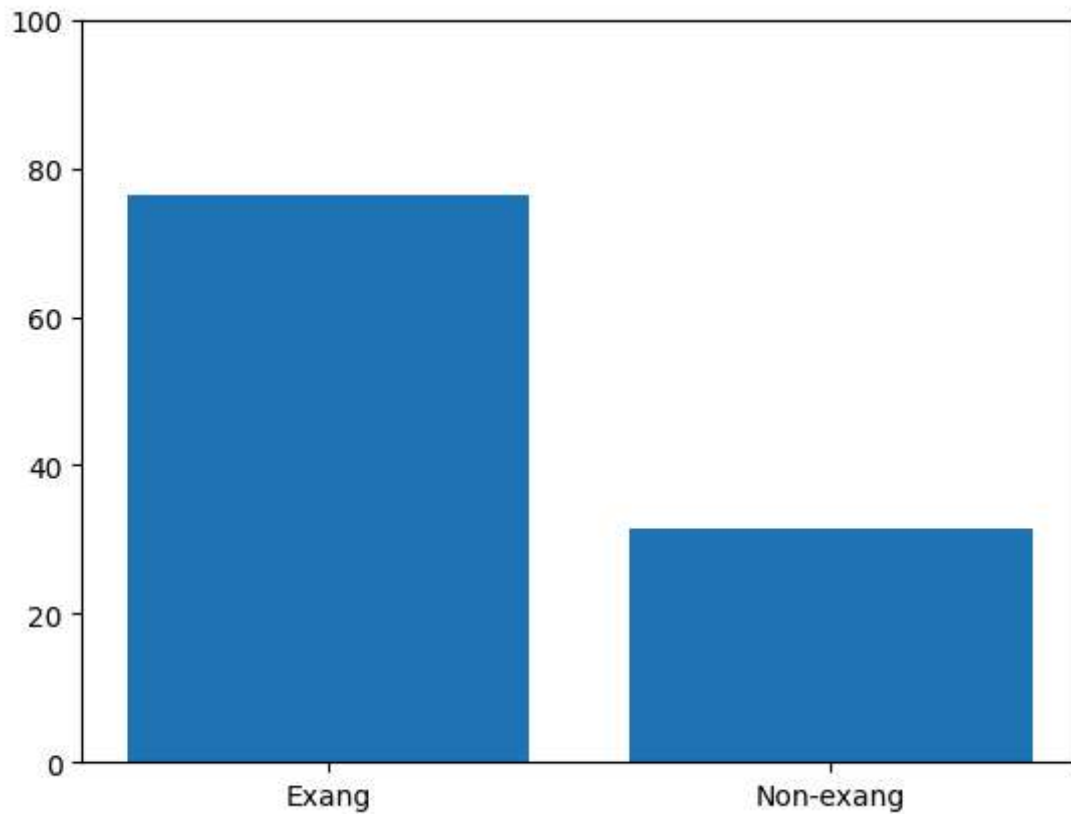


```
In [42]: exang = df[df['exang'] == 1]
nonExang = df[df['exang'] == 0]
exangCount = exang.exang.count()
nonExangCount = nonExang.exang.count()
exangPercent = (exang.condition.sum() / exangCount) * 100
nonExangPercent = (nonExang.condition.sum() / nonExangCount) * 100
print(f"Exang with disease percentage { exangPercent : .2f}")
print(f"Non-exang with disease percentage { nonExangPercent : .2f}")
```

Exang with disease percentage 76.29

Non-exang with disease percentage 31.50

```
In [251]: plt.bar(['Exang', 'Non-exang'], [exangPercent, nonExangPercent])
plt.ylim(0, 100)
plt.show()
```

```
In [260]: illOldPeak = df[df['condition'] == 1].oldpeak
nonIllOldPeak = df[df['condition'] == 0].oldpeak

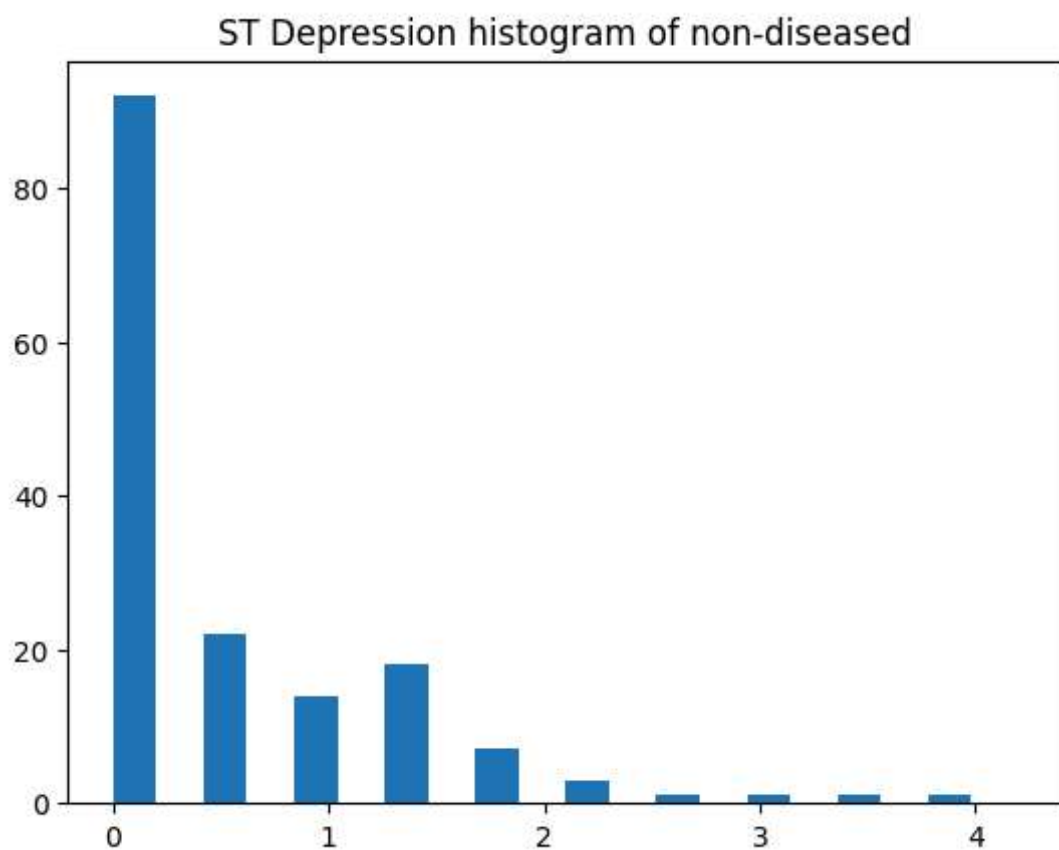
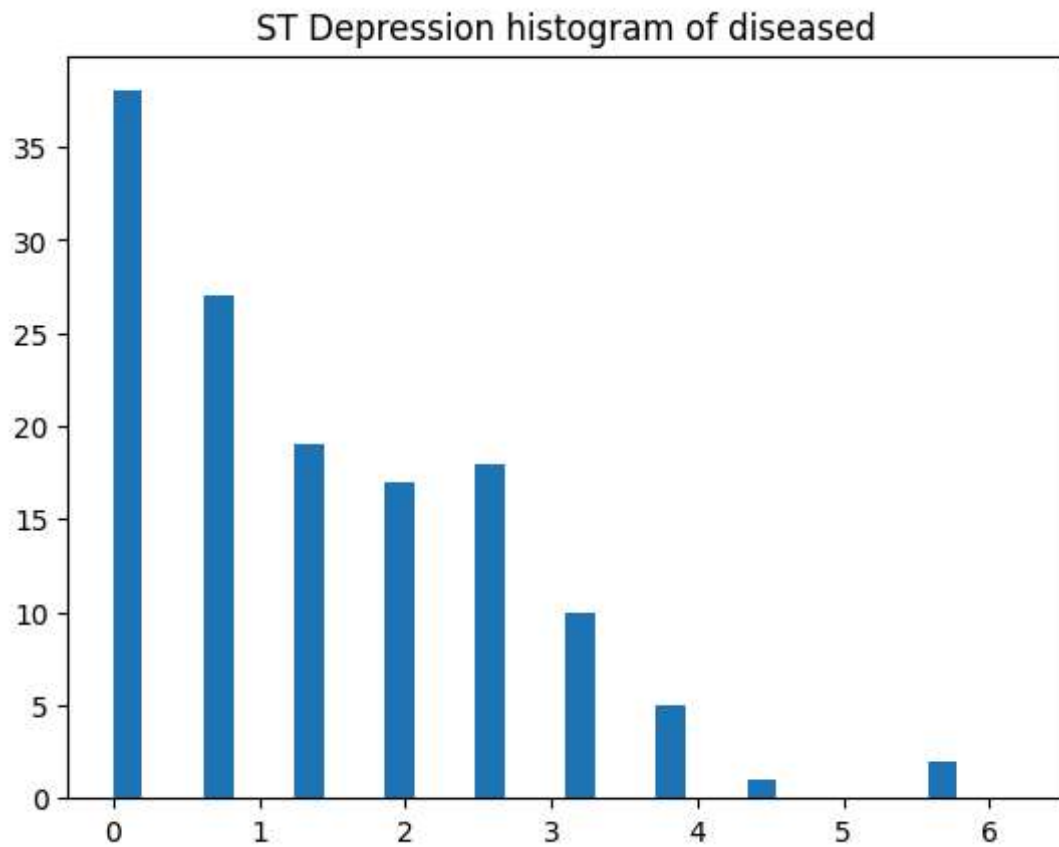
print("Diseased mean of ST Depression ",illOldPeak.mean())
print("Non-diseased mean of ST Depression ",nonIllOldPeak.mean())

plt.hist(illOldPeak, width = 0.2)
plt.title("ST Depression histogram of diseased")
plt.show()

plt.hist(nonIllOldPeak, width = 0.2)
plt.title("ST Depression histogram of non-diseased")
plt.show()
```

Diseased mean of ST Depression 1.5890510948905108

Non-diseased mean of ST Depression 0.5987500000000001

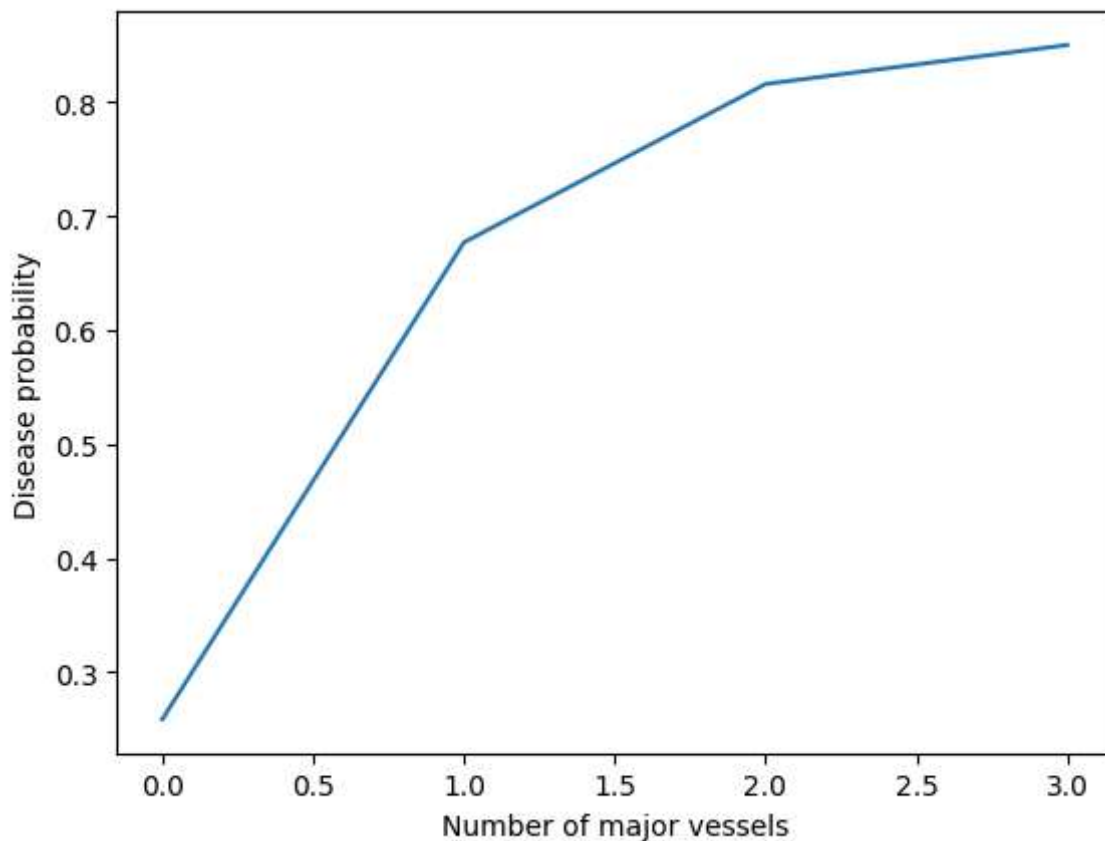


```
In [5]: caGroup = df.groupby('ca').condition
```

```
In [7]: caIll = caGroup.sum()  
caAll = caGroup.count()  
caProbability = caIll/caAll  
caProbability
```

```
Out[7]: ca
0      0.258621
1      0.676923
2      0.815789
3      0.850000
Name: condition, dtype: float64
```

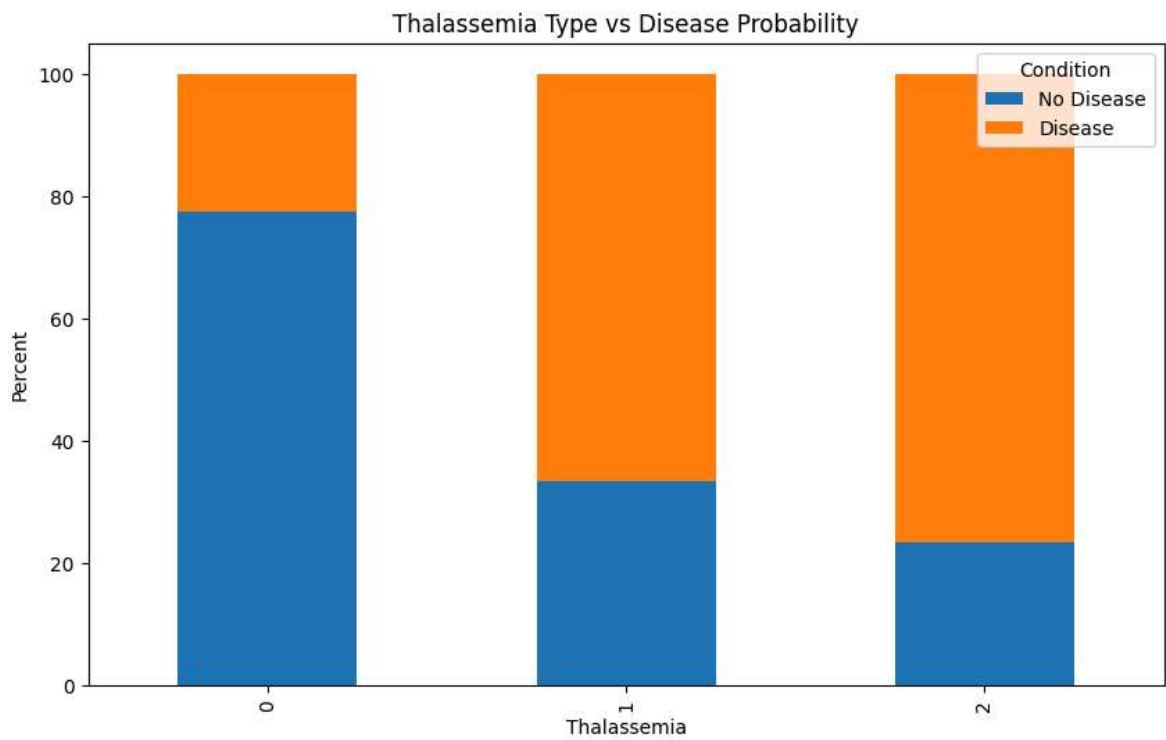
```
In [10]: plt.plot(caProbability.index, caProbability)
plt.xlabel('Number of major vessels')
plt.ylabel('Disease probability')
plt.show()
```



```
In [25]: cross_tab = pd.crosstab(df['thal'], df['condition'], normalize='index') * 100
cross_tab.plot(kind='bar', stacked=True, figsize=(10, 6))

plt.title('Thalassemia Type vs Disease Probability')
plt.xlabel('Thalassemia')
plt.ylabel('Percent')
plt.legend(['No Disease', 'Disease'], title='Condition', loc='upper right')
# plt.show()
# print(pd.crosstab.__doc__)
```

```
Out[25]: <matplotlib.legend.Legend at 0x18862872540>
```

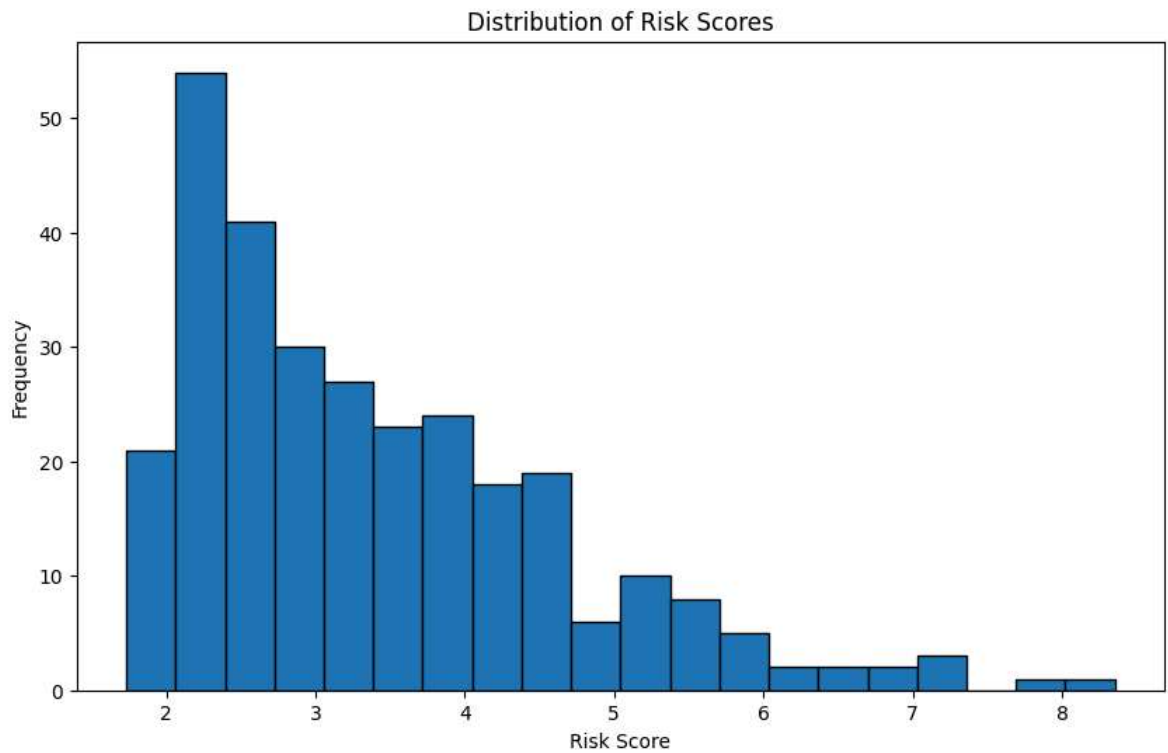


```
In [290]: condition = df['condition'].to_numpy()
patients = (df['age'].to_numpy() > 50) & (df['chol'].to_numpy() > 240) & (df['trestbps'] > 120)
diseasePercentage = np.mean(condition[patients]) * 100
diseasePercentage
```

Out[290]: 66.66666666666666

```
In [296]: df['risk_score'] = (df['chol'] / 200) + (df['trestbps'] / 120) + df['oldpeak']
conditions = [
    (df['risk_score'] <= 2),
    (df['risk_score'] > 2) & (df['risk_score'] <= 4),
    (df['risk_score'] > 4)
]
risk_labels = ['Low Risk', 'Medium Risk', 'High Risk']
df['risk_category'] = np.select(conditions, risk_labels, default='Medium Risk')

plt.figure(figsize=(10, 6))
plt.hist(df['risk_score'], bins=20, edgecolor='black')
plt.title('Distribution of Risk Scores')
plt.xlabel('Risk Score')
plt.ylabel('Frequency')
plt.show()
print(df['risk_category'].value_counts())
```



```

risk_category
Medium Risk ... 200
High Risk ... 81
Low Risk ... 16
Name: count, dtype: int64

```

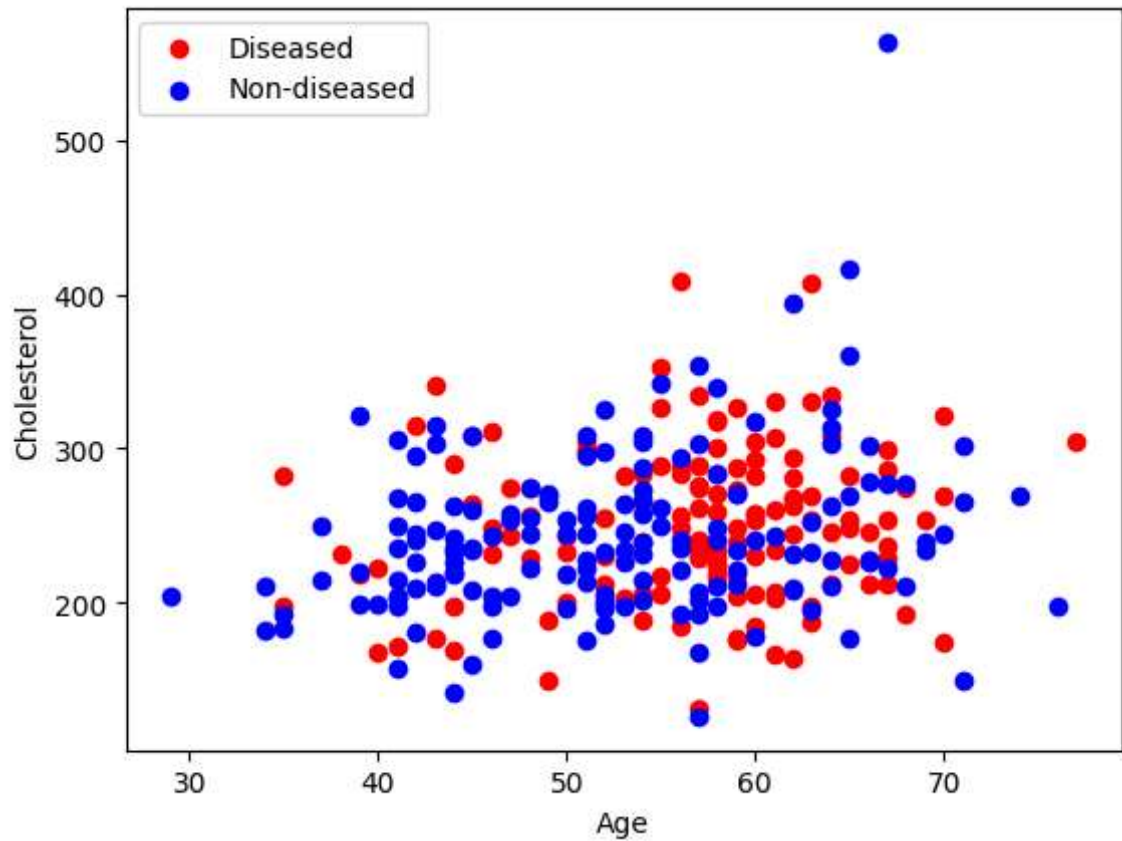
In []:

```

In [39]: # Does cholesterol strongly impact heart disease?
# No
plt.scatter(df['age'][df['condition'] == 1], df['chol'][df['condition'] == 1], co
plt.scatter(df['age'][df['condition'] == 0], df['chol'][df['condition'] == 0], co

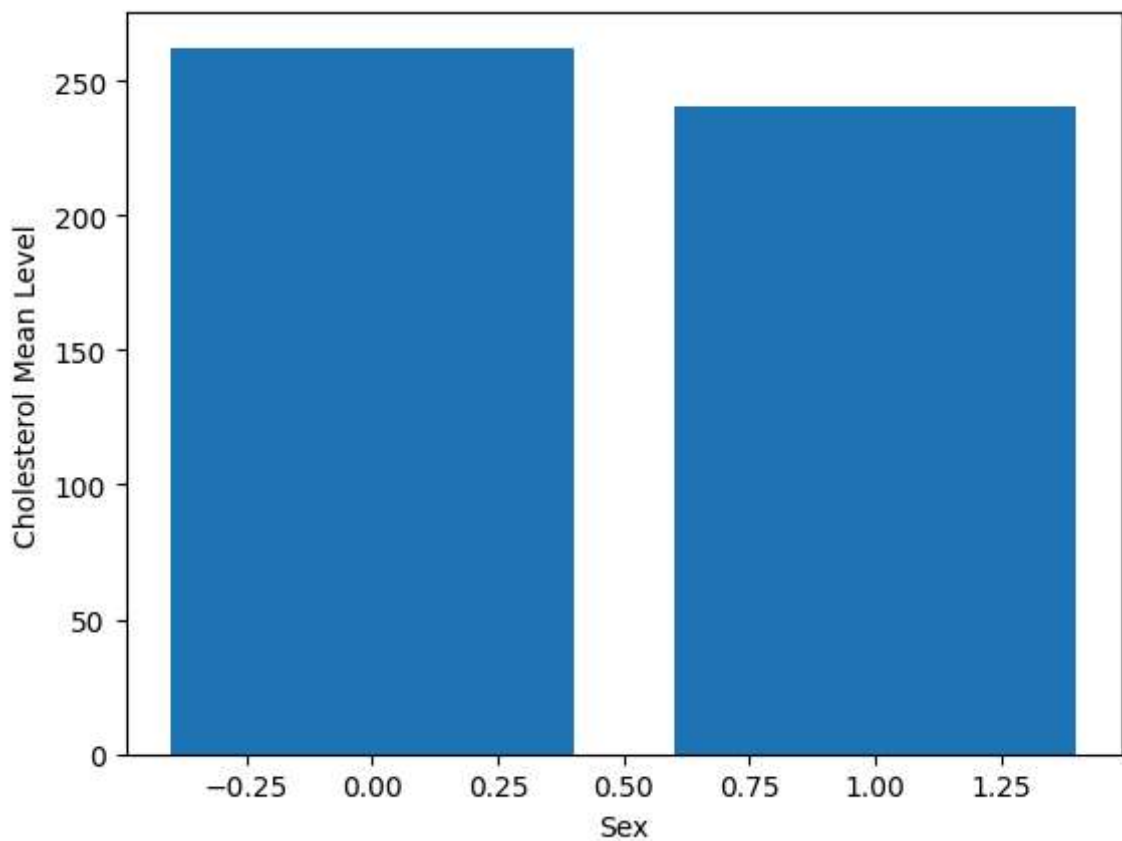
plt.xlabel('Age')
plt.ylabel('Cholesterol')
plt.legend()
plt.show()

```



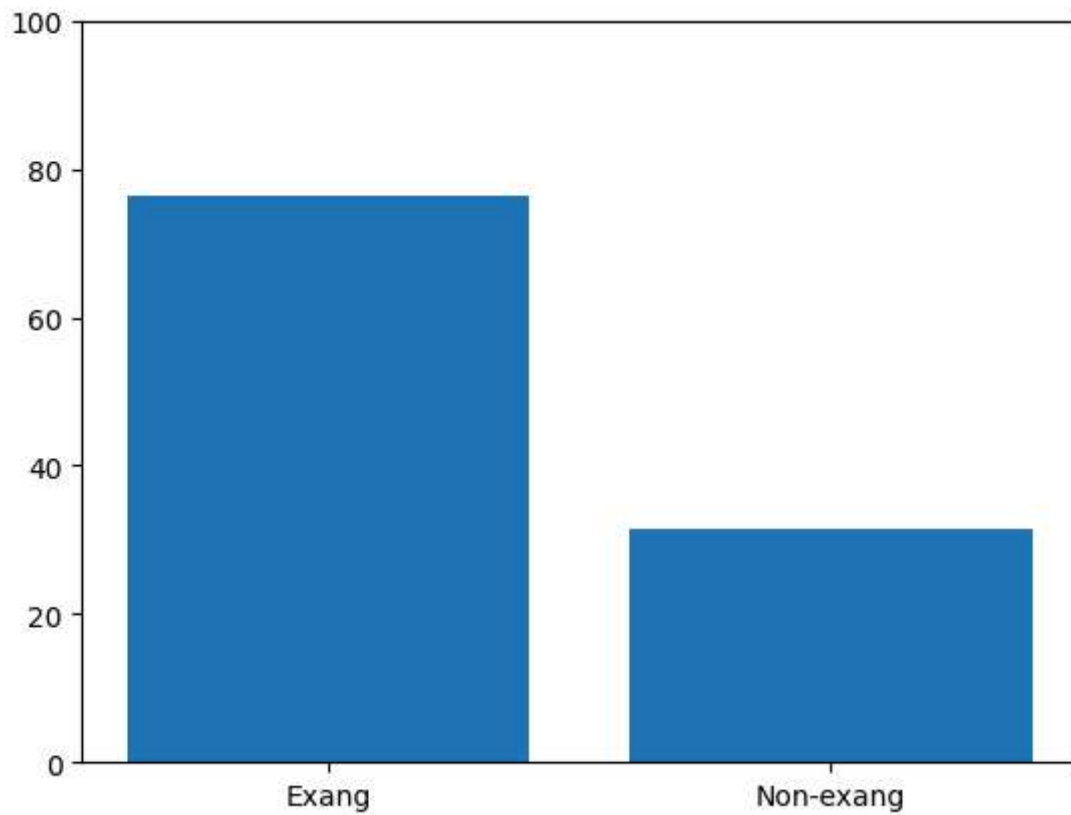
```
In [40]: # Is male population more vulnerable?  
# Not significantly, only slightly more
```

```
plt.bar(sexGroupMeanChol.index, sexGroupMeanChol)  
plt.xlabel("Sex")  
plt.ylabel("Cholesterol Mean Level")  
plt.show()
```



```
In [43]: # Does exercise-induced angina significantly increase risk?
# Yes, almost doubles

plt.bar(['Exang', 'Non-exang'], [exangPercent, nonExangPercent])
plt.ylim(0, 100)
plt.show()
```



```
In [44]: # Which feature has strongest correlation with disease?
# Number of major vessels

df['ca'].corr(df['condition'])
```

Out[44]: 0.4631886250630225